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ON A THEORY OF THE SYZYGETIC RELATIONS OF TWO RATIONAL
INTEGRAL FUNCTIONS, COMPRISING AN APPLICATION TO THE THEORY
OF STURM'S FUNCTIONS, AND THAT OF THE GREATEST ALGEBRAICAL
COMMON MEASURE

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Adopting the language which I have frequently employed elsewhere, I call $\mathfrak{S}_t$ a syzygetic function, or more briefly a conjunctive of f and ϕ , and Λ_t and L_t may be termed the syzygetic factors to $\mathfrak{S}_t$ so considered. If we divide each term of the equation (15) by the allotropic factor (M), we have

$$\frac{\Lambda_t}{M}f - \frac{L_t}{M}\phi = R_t,$$

where R_t is the t th simplified residue to (f, ϕ) ; and if we call $\frac{\Lambda_t}{M} = \tau_t$, and $\frac{L_t}{M} = t_t$, so as to obtain the equation

$$\tau_t f - t_t \phi = R_t, \tag{16}$$

we see that $\frac{\tau_t}{t_t}$, the fraction formed by the component factors to any simplified residue of (f, ϕ) , will be identical in value (although no longer in its separate terms) with one of the corresponding convergents to $\frac{\phi}{f}$, exhibited under the form of an improper continued fraction. I shall in the next section show how, not only the successive simplified residues, but also the component syzygetic factors of each of them, and consequently the successive convergents, may be expressed in terms of the roots of the two given functions.

Since the preceding section was composed the valuable memoir of the lamented Jacobi, entitled “De Eliminatione Variabilis è duabus Equationibus Algebraicis,” Crelle, Vol. xvi., has fallen under my notice. That memoir is restricted to the consideration of two equations of the same degree, and the principal results in this section as regards the Bezoutic square and the allotropic factors applicable to that case will be found contained therein. The mode of treatment however is sufficiently dissimilar to justify this section being preserved unaltered under its original form.

SECTION II.

On the general solution in terms of the roots, of any two given algebraical functions of x , of the syzygetic equation, which connects them with a third function, whose degree in x is given, but whose form is to be determined.

These two forms are only the two extremities of a scale of forms all equally well adapted to express $\mathfrak{S}_t$; for let v and v' be any two integers so taken as to satisfy the equation

$$v + v' = t,$$

and let $R(\overline{\cdots}; \overline{\cdots})$, where the dots denote any quantities whatever, be used to denote a rational function which remains unaltered in value when any two of the quantities under either of the two bars are mutually interchanged, then we may write

$$\mathfrak{S}_t = \Sigma \left\{ \frac{R(\overline{h_{q_1}, h_{q_2} \dots h_{q_v}}; \overline{\eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}}}) \times \left[\begin{array}{c} h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{array} \right]}{\times (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_v})(x - \eta_{\xi_1})(x - \eta_{\xi_2}) \dots (x - \eta_{\xi_{v'}})} \right\}. \quad (19)$$

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For if, as above, we suppose $x = h_a = \eta_\omega$, any term of $\mathfrak{S}_t$ in which $q_1, q_2 \dots q_v$ comprise among them a , or in which $\xi_1, \xi_2 \dots \xi_{v'}$ comprise among them ω , will vanish by virtue of the factors

$$(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_v}) \times (x - \eta_{\xi_1})(x - \eta_{\xi_2}) \dots (x - \eta_{\xi_{v'}});$$

but if neither a nor ω is so comprised, then a must be one of the terms in the complementary series $q_{v+1}, q_{v+2} \dots q_m$, and ω one of the terms in the complementary series $\xi_{v'+1}, \xi_{v'+2} \dots \xi_n$, and therefore one of the quantities $h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m}$ will equal one of the quantities $\eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n}$, and consequently the term of $\mathfrak{S}_t$ in question will vanish by virtue of the factor

$$\left[\begin{array}{c} h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{array} \right]$$

vanishing. In either case therefore every term included within the sign of summation vanishes when $x = h_a = \eta_\omega$, that is, whenever $fx = 0$ and $\phi x = 0$. Hence $\mathfrak{S}_t$, as given by equation (19), will satisfy the syzygetic equation $\tau_t f - t_t \phi + \mathfrak{S}_t = 0$ for all values of v and v' which make $v + v' = t$, and for all symmetrical forms of the function denoted by the symbol $R(\overline{\cdots}; \overline{\cdots})$.

Art. (16). I shall now proceed to show how to assign the arbitrary function whose form is denoted by this symbol in such a manner as to make $\mathfrak{S}_t$ become identical with a simplified residue to f and ϕ . To this end I take for $R(h_{q_1}, h_{q_2} \dots h_{q_v}; \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}})$ the value

$$R = \frac{\left[\begin{array}{c} h_{q_1}, h_{q_2} \dots h_{q_v} \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \end{array} \right]}{\left[\begin{array}{c} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{array} \right]} \times \left[\begin{array}{c} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{array} \right]; \quad (20)$$

we shall then have

$$\mathfrak{S}_t = \Sigma \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \end{bmatrix} \times \begin{bmatrix} h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{bmatrix} \times \begin{bmatrix} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{bmatrix}} \times \{(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_v})(x - \eta_{\xi_1})(x - \eta_{\xi_2}) \dots (x - \eta_{\xi_{v'}})\}. \quad (21)$$

I shall first show this sum of fractions is in substance an integral function of the quantities $h_1, h_2 \dots h_m; \eta_1, \eta_2 \dots \eta_n$. For greater conciseness write in general $x - h = E, x - \eta = H$; we have then, since

$$h - \eta = H - E, \quad h_{q_r} - h_{q_s} = E_{q_s} - E_{q_r}, \quad \eta_{\xi_r} - \eta_{\xi_s} = H_{\xi_s} - H_{\xi_r},$$

$$\mathfrak{S}_t = \Sigma \frac{\begin{bmatrix} H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \\ E_{q_1}, E_{q_2} \dots E_{q_v} \end{bmatrix} \times \begin{bmatrix} H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \\ E_{q_{v+1}}, E_{q_{v+2}} \dots E_{q_m} \end{bmatrix}}{\begin{bmatrix} E_{q_{v+1}}, E_{q_{v+2}} \dots E_{q_m} \\ E_{q_1}, E_{q_2} \dots E_{q_v} \end{bmatrix} \times \begin{bmatrix} H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \\ H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \end{bmatrix}} E_{q_1} \dots E_{q_v} H_{\xi_1} \dots H_{\xi_{v'}}. \quad (22)$$

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On reducing the fractions contained within the sign of summation to a common denominator, $\mathfrak{S}_t$ will take the form $\frac{N}{D \cdot \Delta}$, where D will be the product of the $\frac{1}{2}m(m-1)$ differences of $E_1, E_2 \dots E_m$ subtracted each from each, and Δ the corresponding product of the differences *inter se* of $H_1, H_2 \dots H_n$. Hence, unless the sum in question is an integral function of the E 's and H 's it will become infinite when any two of the E series, or any two of the H series of quantities are made equal. Suppose now $E_1 = E_2$; the terms in (22) which contain $E_1 - E_2$ in the denominator will evidently group themselves into pairs of the respective forms,

$$\frac{(E_1 E_{q_3} \dots E_{q_v}) \times (H_{\xi_1} H_{\xi_2} \dots H_{\xi_{v'}}) \times \begin{bmatrix} E_1, E_{q_3} \dots E_{q_v} \\ H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \end{bmatrix} \times \begin{bmatrix} E_2, E_{q_{v+2}} \dots E_{q_m} \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}}{\begin{bmatrix} E_1, E_{q_3} \dots E_{q_v} \\ E_2, E_{q_{v+2}} \dots E_{q_m} \end{bmatrix} \times \begin{bmatrix} H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}}$$

and

$$\frac{(E_2 E_{q_3} \dots E_{q_v}) \times (H_{\xi_1} H_{\xi_2} \dots H_{\xi_{v'}}) \times \begin{bmatrix} E_2, E_{q_3} \dots E_{q_v} \\ H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \end{bmatrix} \times \begin{bmatrix} E_1, E_{q_{v+2}} \dots E_{q_m} \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}}{\begin{bmatrix} E_2, E_{q_3} \dots E_{q_v} \\ E_1, E_{q_{v+2}} \dots E_{q_m} \end{bmatrix} \times \begin{bmatrix} H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}};$$

the sum of this pair of terms will be of the form

$$\frac{P}{Q} \left\{ \frac{E_1}{E_1 - E_2} \frac{\begin{bmatrix} E_1 \\ H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \end{bmatrix} \times \begin{bmatrix} E_2 \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}}{\begin{bmatrix} E_1 \\ E_{q_{v+1}}, E_{q_{v+2}} \dots E_{q_m} \end{bmatrix}} + \frac{E_2}{E_2 - E_1} \frac{\begin{bmatrix} E_2 \\ H_{\xi_1}, H_{\xi_2} \dots H_{\xi_{v'}} \end{bmatrix} \times \begin{bmatrix} E_1 \\ H_{\xi_{v'+1}}, H_{\xi_{v'+2}} \dots H_{\xi_n} \end{bmatrix}}{\begin{bmatrix} E_2 \\ E_{q_{v+1}}, E_{q_{v+2}} \dots E_{q_m} \end{bmatrix}} \right\},$$

where Q , it may be observed, does not contain $H_1 - H_2$, so that $\frac{P}{Q}$ remains finite when $H_1 = H_2$.

The above pair of terms together make up a sum of the form

$$\frac{P}{Q} \frac{1}{E_1 - E_2} \frac{\phi(E_1, E_2)\psi E_2 - \phi(E_2, E_1)\psi E_1}{\psi E_1 \times \psi E_2},$$

which, as the numerator of the third factor vanishes when $E_1 = E_2$, remains finite on that supposition. Hence the whole sum of terms in (22) which is p. 463

made up of such pairs of terms, and of other terms in which $E_1 - E_2$ does not enter, remains finite when $E_1 - E_2 = 0$, and therefore generally when $D = 0$, and similarly when $H_1 - H_2 = 0$, and therefore also when $\Delta = 0$; hence the expression for $\mathfrak{S}_t$ in (22) is an integral function of the E and H series of quantities, as was to be proved.

Art. (17). Let us now proceed to determine the dimensions of the coefficient of x^t , the highest power of x in this value of $\mathfrak{S}_t$, when supposed to be expressed under the form of an integral function (as it has been proved to be capable of being expressed) of $h_1, h_2 \dots h_m; \eta_1, \eta_2 \dots \eta_n; x$.

This coefficient is the sum of fractions the numerators of each of which consist of two factors, which are respectively of $v \times v'$ and of $(m-v) \times (n-v')$ dimensions in respect of the two sets of roots taken conjointly, and the denominators of two factors respectively of $v(m-v)$ and $v'(n-v')$ dimensions in respect of the same.

Consequently, the exponent of the total dimensions of the coefficient in question

$$= vv' + (m-v)(n-v') - v(m-v) - v'(n-v') = (m-v-v') \times (n-v-v') = (m -$$

and thus is seen to depend only on the degree t in x of $\mathfrak{S}_t$, and not upon the mode of partitioning t into two parts v and v' , for the purpose of representing $\mathfrak{S}_t$ by means of formula (19).

Art. (18). I shall now demonstrate that every form in this scale (to a numerical factor *près*) is identical with a simplified residue to f, ϕ , of the same degree t in x . Any such simplified residue is, like $\mathfrak{S}_t$, a syzygetic function, or to use a briefer form of speech a conjunctive of f, ϕ ; and if we agree to understand by the "weight" of any function of the coefficients of f and ϕ its joint dimensions in respect of the roots of f and ϕ combined, I shall prove,—first, that any simplified residue of f and ϕ of a given degree in x is that conjunctive, whose weight in respect of the roots of f and ϕ is less than the weight of any other such conjunctive; and second, that $\mathfrak{S}_t$, as determined above (in equation 22), is of the same weight as the simplified residue, and can therefore only differ from it by some numerical factor. For the purpose of comparison of weights, it will of course be sufficient to confine our attention to the coefficients of the highest power in x (or any other, the same for each) of the forms whose weights are to be compared.

Suppose f to be of m dimensions, and ϕ to be of n dimensions in x ; and let $m = n + e$.

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Suppose

$$\begin{aligned}\Lambda f + L\phi &= Ax^t + Bx^{t-1} + \cdots + K, \\ \Lambda &= \lambda_0 x^q + \lambda_1 x^{q-1} + \cdots + \lambda_q, \\ L &= l_0 x^{q+e} + l_1 x^{q+e-1} + \cdots + l_{q+e},\end{aligned}\tag{23}$$

the number of homogeneous equations to be satisfied by the $q + 1$ quantities $\lambda_0, \lambda_1 \dots \lambda_q$, and the $q + e + 1$ quantities $l_0, l_1 \dots l_{q+e}$ will be $m + q - t$, and therefore $q + 1$ and $q + e + 1$ taken together must be not less than $m + q - t + 1$, that is $2q + e + 2$ must be not less than $q + m - t + 1$, that is q not less than $m - t - e - 1$; and if this inequality be satisfied $2q + e + 2 - (q + m - t + 1) + 1$, that is $q + t + e - m + 2$ will be the number of arbitrary constants entering into the solution of equation (23).

If q be greater than $(n - 1)$, let $q = (n - 1) + t$; and let

$$\begin{aligned}(\Lambda) &= \lambda_0 x^{n-1} + \lambda_1 x^{n-2} + \cdots + \lambda_{n-1}, \\ (L) &= l_0 x^{n+e-1} + l_1 x^{n+e-2} + \cdots + l_{e+n-1};\end{aligned}$$

and let $(\Lambda), (L)$ be so taken as to satisfy the equation

$$(\Lambda)f + (L)\phi = Ax^t + Bx^{t-1} + \cdots + K;$$

and make

$$\begin{aligned}\Xi &= (\Lambda) + (f + gx + \cdots + hx^{t-1})\phi, \\ X &= (L) - (f + gx + \cdots + hx^{t-1})f,\end{aligned}$$

$f, g \dots h$ being arbitrary constants; then

$$\Xi f + X\phi = (\Lambda)f + (L)\phi = Ax^t + Bx^{t-1} + \cdots + K.$$

Now the total number of arbitrary constants in the system (Λ) and (L) will be $n - 1 + t + e - m + 2$, that is $t + 1$; hence the total number of arbitrary constants in Ξ and X will be $t + 1 + t$, that is $q - n + t + 2$, which is equal to $q + t + e - m + 2$, the number of arbitrary constants in the most general values of Λ and L . Hence $(\Lambda = \Xi, L = X)$ is the general solution of the equation $\Lambda f + L\phi = Ax^t + Bx^{t-1} + \cdots + K$; and consequently the most general form of $Ax^t + Bx^{t-1} + \cdots + K$, which is evidently independent of the (t) arbitrary quantities $f, g \dots h$, will contain the same number of arbitrary constants as enter into the system (Λ) and (L) , that is $t + 1$.

Art. (19). Let us now begin with the case of greater simplicity when $m = n$, that is $e = 0$; and let us revert to the system of equations marked (10) in Section I., in which U and V are to be replaced by f and ϕ .

First, let $t = n - 1$, then $t + 1$, the number of arbitrary quantities, in the conjunctive, is n .

From the system of equations (10) we have, for all values of $\rho_1, \rho_2, \rho_3 \dots \rho_n$,

$$\begin{aligned} (\rho_1 Q_0 + \rho_2 Q_1 + \dots + \rho_n Q_{n-1})f - (\rho_1 P_0 + \rho_2 P_1 + \dots + \rho_n P_{n-1})\phi \\ = (\rho_1 K_1 + \rho_{21} K_1 + \dots + \rho_{nn-1} K_1)x^{n-1} + \&c., \end{aligned}$$

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and consequently the most general value of $\mathfrak{S}_{n-1}$ in the equation

$$\tau_{n-1}f - t_{n-1}\phi + \mathfrak{S}_{n-1} = 0,$$

where

$$\mathfrak{S}_{n-1} = Ax^{n-1} + Bx^{n-2} + \dots + L,$$

will be obtained by making

$$\tau_{n-1} = \rho_1 Q_0 + \rho_2 Q_1 + \dots + \rho_n Q_{n-1},$$

$$t_{n-1} = -\rho_1 P_0 - \rho_2 P_1 - \dots - \rho_n P_{n-1},$$

which solution contains n , that is the proper number of arbitrary constants.

Again, if $t = n - 2$, $t + 1 = n - 1$, which will therefore be the number of arbitrary constants in the most general value of $\mathfrak{S}_{n-2}$ in the equation

$$\tau_{n-2}f - t_{n-2}\phi + \mathfrak{S}_{n-2} = 0.$$

This most general value of $\mathfrak{S}_{n-2}$ is therefore found by making

$$\tau_{n-2} = \rho'_1 Q_0 + \rho'_2 Q_1 + \dots + \rho'_n Q_{n-1},$$

$$t_{n-2} = -\rho'_1 P_0 - \rho'_2 P_1 - \dots - \rho'_n P_{n-1},$$

where $\rho'_1, \rho'_2 \dots \rho'_n$ are no longer entirely independent, but subject to the equation

$$\rho'_1 K_1 + \rho'_{21} K_1 + \dots + \rho'_{nn-1} K_1 = 0,$$

so as to leave $(n - 1)$ constants arbitrary.

We thus obtain

$$\mathfrak{S}_{n-2} = (\rho'_1 K_2 + \rho'_{21} K_2 + \dots + \rho'_{nn-1} K_2)x^{n-2} + \&c.$$

In like manner, and for the same reasons, the most general values of $\mathfrak{S}_{n-3}$ in the equation

$$\tau_{n-3}f - t_{n-3}\phi + \mathfrak{S}_{n-3} = 0,$$

will be found by making

$$\tau_{n-3} = \rho''_1 Q_0 + \rho''_2 Q_1 + \dots + \rho''_n Q_{n-1},$$

$$t_{n-3} = -\rho''_1 P_0 - \rho''_2 P_1 - \dots - \rho''_n P_{n-1},$$

where $\rho_1'', \rho_2'' \dots \rho_n''$ are subject to satisfying the two equations

$$\begin{aligned}\rho_1'' K_1 + \rho_{21}'' K_1 + \dots + \rho_{nn-1}'' K_1 &= 0, \\ \rho_1'' K_2 + \rho_{21}'' K_2 + \dots + \rho_{nn-1}'' K_2 &= 0,\end{aligned}$$

so as to leave $(n - 2)$ constants arbitrary; and we thus obtain

$$\mathfrak{S}_{n-3} = (\rho_1'' K_3 + \rho_{21}'' K_3 + \dots + \rho_{nn-1}'' K_3) x^{n-3} + \&c.,$$

and so on, the number of independent arbitrary constants in $\mathfrak{S}$ decreasing (as it ought) each time by one unit as the degree of $\mathfrak{S}$ descends, until finally, if $\tau_0 f - t_0 \phi + \mathfrak{S}_0 = 0$, $\mathfrak{S}_0$ being a constant, the general value of $\mathfrak{S}_0$ is found by making

$$\begin{aligned}\tau_0 &= (\rho_1) Q_0 + (\rho_2) Q_1 + \dots + (\rho_n) Q_{n-1}, \\ t_0 &= -(\rho_1) P_0 - (\rho_2) P_1 - \dots - (\rho_n) P_{n-1},\end{aligned}$$

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where $(\rho_1), (\rho_2) \dots (\rho_n)$ are subject to satisfy the $(n - 1)$ equations

$$\begin{aligned}(\rho_1) K_1 + \&c. &= 0, \\ (\rho_1) K_2 + \&c. &= 0, \\ &\dots\dots\dots \\ (\rho_1) K_{n-1} + \&c. &= 0,\end{aligned}$$

which gives

$$\mathfrak{S}_0 = K_n(\rho_1) + {}_1K_n(\rho_2) + \dots + {}_{n-1}K_n(\rho_n).$$

Now evidently the lowest weight in respect to the roots of U and V that can be given to $(\rho_1 K_1 + \rho_{21} K_1 + \dots + \rho_{nn-1} K_1) x^{n-1} + \&c.$, when the multipliers $\rho_1, \rho_2 \dots \rho_n$ are absolutely independent, is found by taking

$$\rho_1 = 1, \quad \rho_2 = 0, \quad \rho_3 = 0 \dots \rho_n = 0,$$

which makes the weight of the leading coefficient in $\mathfrak{S}_{n-1}$, the same as that of K_1 , that is 1.

Again, when one equation,

$$\rho_1' K_1 + \rho_{21}' K_1 + \dots + \rho_{nn-1}' K_1 = 0,$$

exists between the (ρ) 's, the lowest weight will be found by making

$$\rho_1' = {}_1K_1, \quad \rho_2' = -K_1, \quad \rho_3' = 0, \quad \rho_4' = 0 \dots \rho_n' = 0,$$

which makes the weight of the leading coefficient in $\mathfrak{S}_{n-2}$ depend on

$${}_1K_1 K_2 - K_{11} K_2,$$

which is of the weight $1 + 3$, that is 4 , in respect of the roots of f and ϕ .

Similarly, $\mathfrak{S}_{n-3}$ will have its lowest weight when its leading coefficient is the determinant

$$\begin{vmatrix} K_1 & K_2 & K_3 \\ {}_1K_1 & {}_1K_2 & {}_1K_3 \\ {}_2K_1 & {}_2K_2 & {}_2K_3 \end{vmatrix},$$

the weight of which is $1 + 3 + 5 = 9$; and finally, the lowest weighted value of $\mathfrak{S}_0$ is the determinant represented by the complete Bezoutian square; the weight in general of $\mathfrak{S}_{n-t}$ being $1 + 3 + \dots + (2t - 1)$, that is t^2 , or which is the same thing otherwise expressed, the weight of the leading coefficient of the lowest-weighted conjunctive of f and ϕ of the degree t in x is $(n - t)(m - t)^1$. It will of course have been seen in the foregoing demonstration, that the weight of ${}_rK_s$ [which means $\Sigma(a_rb_s - a_sb_r)$, a_r, a_s being the coefficients of x^{n-r}, x^{n-s} in f , and b_r, b_s of the same in ϕ] has been correctly taken to be $r + s$ in respect of the roots of f and ϕ conjoined.

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Art. (20). If now we proceed in like manner with the general case of $m = n + e$, it may be shown, in precisely the same way as in the preceding article, that the most general value of any conjunctive of f and ϕ will be a linear function of e functions,

$$\begin{aligned} & x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n, \\ & x^{n+1} + a_1x^n + a_2x^{n-1} + \dots + a_nx, \\ & x^{n+2} + a_1x^{n+1} + a_2x^n + \dots + a_nx^2, \\ & \dots\dots\dots, \\ & x^{m-1} + a_1x^{m-2} + \&c. \quad + \quad a_nx^{e-1}, \end{aligned}$$

and of the n functions,

$$\begin{aligned} & K_1x^{n-1} + \quad K_2x^{n-2} + \quad \dots + \quad K_n, \\ & {}_1K_1x^{n-1} + \quad {}_1K_2x^{n-2} + \quad \dots + \quad {}_1K_n, \\ & \quad \&c. \quad \quad \quad \&c. \\ & {}_{n-1}K_1x^{n-1} + \quad {}_{n-1}K_2x^{n-2} + \quad \dots + \quad {}_{n-1}K_n, \end{aligned}$$

and that consequently, if the degree of such conjunctive in x be $(n - i)$, it will be of the lowest weight when it is a linear function of the entire e upper set of functions, and i of the lower set; and consequently, the coefficient of the highest power of x in such conjunctive will be the determinant

$$\begin{vmatrix} K_1 & K_2 & K_3 & \dots & K_i & \dots K_{i+e} \\ {}_1K_1 & {}_1K_2 & {}_1K_3 & \dots & {}_1K_i & \dots {}_1K_{i+e} \\ {}_2K_1 & {}_2K_2 & {}_2K_3 & \dots & {}_2K_i & \dots {}_2K_{i+e} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ {}_{i-1}K_1 & {}_{i-1}K_2 & {}_{i-1}K_3 & \dots & {}_{i-1}K_i & \dots {}_{i-1}K_{i+e} \\ 1 & a_1 & a_2 & \dots a_{i-1} & a_i & \dots a_{i+e-1} \\ 0 & 1 & a_1 & \dots a_{i-1} & a_i & \dots a_{i+e-2} \\ 0 & 0 & 1 & \dots a_{i-2} & a_i & \dots a_{i+e-3} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & \dots a_i \end{vmatrix},$$

¹ n and m are supposed equal and $t = n - i$.

the weight of which is evidently that of

$$K_1 \times K_2 \times K_3 \times \cdots \times K_{i-1} \times (a_i)^e,$$

that is

$$1 + 3 + 5 + \cdots + (2i - 1) + ei,$$

that is $i^2 + ei$, or $i(e + i)$, which is $(n - t)(m - t)$ if $t = n - i$.

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Hence the weight of the leading coefficient in the lowest-weighted conjunctive of f and ϕ of the degree t in x is $(m - t)(n - t)$, m being the degree of f and n of ϕ .

From this we infer that any conjunctive of f and ϕ of the degree t , of which the leading coefficient is of the weight $(m - t)(n - t)$, all the coefficients being of course understood to be integral functions of the roots of f and ϕ , must, to a numerical factor *près*, be equivalent to any other of the same weight; and furthermore, any supposed function of x of the t th degree which possesses the property characteristic of a conjunctive of vanishing when f and ϕ vanish simultaneously, but of which the weight of the leading coefficient would be *less* than $(m - t)(n - t)$, must be a mere nugatory form and have all its terms identically zero².

Art. (21). We have previously shown, Art. 16, that $\mathfrak{S}$, as defined by equation (21), is an integral function of the roots f and ϕ , and vanishes when f and ϕ vanish. Moreover, its weight in the roots has been proved to be $(m - t)(n - t)$, and consequently, if by way of distinguishing the several forms of $\mathfrak{S}_t$ we name that one where t in the equation above cited is supposed to be divided into two parts, v and v' , $\mathfrak{S}_{v,v'}$, we have for all values of v and v' , such that $v + v'$ is not greater than n , $\mathfrak{S}_{v,v'}$ to a constant numerical factor *près* identical with the $(v + v')$ th simplified residue to (f, ϕ) , so that the form of $\mathfrak{S}_{v,v'}$ depends only upon the value of $v + v'$.

Art. (22). It must be well borne in mind that this permanency of the value of $\mathfrak{S}_{v,v'}$ for different values of v has only been established for the case where t can be the degree of a residue to f and ϕ , that is to say, when t is less than the lesser of the two indices m and n . When t does not satisfy this condition of inequality, the theorem ceases to be true. It is clear that when $m = n$ and $v + v' = m = n$, $\mathfrak{S}_{v,v'}$, which always remains a conjunctive of f and ϕ , can only be a numerical linear function of f and ϕ ; and I have ascertained when $m = n$ on giving to v and v' the respective values successively $(0, n), (1, n - 1), (2, (n - 2)) \dots (n, 0)$ that

$$\mathfrak{S}_{0,n} = f, \quad \mathfrak{S}_{1,n-1} = (n-1)f + \phi, \quad \mathfrak{S}_{2,n-2} = \frac{(n-1)(n-2)}{1 \cdot 2} f + (n-1)\phi, \dots$$

²And more generally it admits of being demonstrated by precisely the same course of reasoning, that the number of arbitrary parameters in a conjunctive of the degree t , and of the weight $(m - t)(n - t) + e$ in the roots, cannot (abstraction being supposed to be made of an arbitrary numerical multiplier) exceed the number e .

$$\mathfrak{S}_{n-1,1} = f + (n-1)\phi, \quad \mathfrak{S}_{n,0} = \phi.$$

Thus, by way of a simple example, let

$$f = x^2 + ax + b = (x - h_1)(x - h_2),$$

$$\phi = x^2 + \alpha x + \beta = (x - k_1)(x - k_2),$$

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$$\mathfrak{S}_{0,2} = (x - h_1)(x - h_2) \left(\frac{\begin{bmatrix} h_1 h_2 \\ \dots \\ h_1 h_2 \end{bmatrix}}{\begin{bmatrix} h_1 h_2 \\ \dots \\ h_1 h_2 \end{bmatrix}} \times \frac{\begin{bmatrix} \dots \\ k_1 k_2 \\ \dots \\ k_1 k_2 \end{bmatrix}}{\begin{bmatrix} \dots \\ k_1 k_2 \\ \dots \\ k_1 k_2 \end{bmatrix}} \right) = (x - h_1)(x - h_2) = f,$$

$$\begin{aligned} \mathfrak{S}_{1,1} &= \Sigma(x - h_1)(x - k_1) \frac{\begin{bmatrix} h_1 \\ k_1 \end{bmatrix}}{\begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \times \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}} \\ &= \Sigma \frac{x - h_1}{h_1 - h_2} \frac{x - k_1}{k_1 - k_2} \{(h_1 - k_1)(h_2 - k_2)\}, \end{aligned}$$

that is

$$\begin{aligned} \mathfrak{S}_{1,1} &= \Sigma \frac{x - h_1}{h_1 - h_2} \left[\frac{1}{k_1 - k_2} \{(x - k_1)(h_1 - k_1)(h_2 - k_2) \right. \\ &\quad \left. - (x - k_2)(h_1 - k_2)(h_2 - k_1)\} \right] \\ &= \Sigma \frac{x - h_1}{h_1 - h_2} \{(h_1 - h_2)x + [(k_1 + k_2)h_2 - (h_1 h_2 + k_1 k_2)]\} \\ &= (x - h_1)x + (x - h_2)x - (k_1 + k_2)x + (h_1 h_2 + k_1 k_2) \\ &= \{x^2 - (h_1 + h_2)x + h_1 h_2\} + \{x^2 - (k_1 + k_2)x + k_1 k_2\} \\ &= (x^2 + ax + b) + (x^2 + \alpha x + \beta) \\ &= f + \phi; \end{aligned}$$

so we find also $\mathfrak{S}_{2,0} = \phi$.

Art. (23). The expression $\mathfrak{S}_{v,v'}$, which is universally a conjunctive of f and ϕ , continues algebraically interpretable so long as $v + v'$ has any value intermediate between 0 and $m + n$; when $v + v' = 0$ we must of course have $v = 0$ and $v' = 0$, and $\mathfrak{S}_{0,0}$ becomes the resultant of f and ϕ ; when $v + v' = m + n$ we must also have the unique solution $v = m$ and $v' = n$, and $\mathfrak{S}_{m,n}$ becomes necessarily $f \times \phi$, which we thus see stands in a sort of antithetical relation to the resultant of f and ϕ , say (f, ϕ) . Nor is it without interest to remark that $f \times \phi = 0$ implies that a factor of f or else of ϕ is zero; and $(f, \phi) = 0$ implies that if a factor of the one

of the functions is zero, so also is a factor of the other, that is that a factor of each or of neither is zero. As t increases from 0 to n or decreases from $m + n$ to $m - 1$, the number of solutions of the equation $v + v' = t$ in the one case, and the number of admissible solutions of the equation $v + v' = t$ in the other case, which is subject to the condition that v must not exceed n , continues to increase by a unit at each step; there being thus $n + 1$ different forms $\mathfrak{S}_{v,v'}$ when $v + v' = n$, and the same number when $v + v' = m - 1$. For all values of t intermediate between n and $(m - 1)$ (both taken exclusively) it is very remarkable that $\mathfrak{S}_{v,v'}$ will vanish, as I proceed to demonstrate.

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Art. (24). The weight of the coefficient of the highest power of $\mathfrak{S}_{v,v'}$ ($v + v'$ being equal to t) is $(m - t)(n - t)$, and consequently, when t is greater than n , and less than m , $\mathfrak{S}_{v,v'}$ would contain fractional functions of the roots of f and ϕ , if there were in it a power x^t , but $\mathfrak{S}_{v,v'}$ has been proved to be always an integer function of the roots. Hence the coefficient of x^t will be zero, and so more generally the first power of x in $\mathfrak{S}_{v,v'}$ of which the coefficient is not zero, will be $x^{t-\omega}$, subject to the condition (since evidently the weight of the several coefficients goes on increasing by units as the degree of the terms in x decreases by the same) that ω be not less than $(m - t)(t - n)$; let then $\omega = (m - t)(t - n)$, $\mathfrak{S}_{v,v'}$ becomes of the form $Ax^{t-\omega} + Bx^{t-\omega-1} + \&c.$, where A is of zero dimensions; but this is impossible if $t - \omega < n$, for then $Ax^{t-\omega} + \&c.$ is a conjunctive of weight lower than the lowest-weighted simplified residue of the degree $t - \omega$. Hence ω is not greater than $t - n$, that is $(m - t)(t - n)$ is not greater than $t - n$, that is $m - t$ cannot be greater than 1, that is t when intermediate between m and n cannot be less than $m - 1$, otherwise $\mathfrak{S}_{v,v'}$ will vanish identically. Moreover, when $t = m - 1$, $\omega = t - n$, and $t - \omega = n$, and accordingly $\mathfrak{S}_{v,m-1-v}$ is not merely, as we might know, *à priori* an algebraical, but more simply a numerical multiple of ϕ for all values of v . The same is of course true also, m being greater than n , for every form $\mathfrak{S}_{n,m-n}$, since this is always a conjunctive of f and ϕ of which the former is of a degree higher than the $\mathfrak{S}$ in question, so that the multiplier of f in this conjunctive must be zero³.

Art. (25). To enter into a further or more detailed examination of the values assumed by $\mathfrak{S}_{v,v'}$ for the most general values of m, n, t , would be to transcend the limits I have proposed to myself in drawing up the present memoir. What we have established is, that to every form of $\mathfrak{S}_{v,t-v}$ appertaining to a value of t between 0 and n , there is a sort of conjugate form for which t lies between $m + n$ and m ; that for $t = m - 1$ or $t = n$, $\mathfrak{S}_{v,t-v}$ becomes a numerical multiplier of ϕ ; and that when t lies in the intermediate region between n and $m - 1$, $\mathfrak{S}_{v,t-v}$ vanishes for all values of v . I pause only for a moment to put together for the purpose of comparison the forms corresponding to t and to $m + n - t$. By Art. 16,

³It thus appears that if the indices m and n do not differ by at least 3 units, $\mathfrak{S}$ will have an actual quantitative existence for all values of t between 0 and $m + n$; or in other words, the failure in the quantitative existence of the forms $\mathfrak{S}_t$ only begins to show itself when this difference is 3; thus if $m = n + 3$, $\mathfrak{S}_n$ exists, and $\mathfrak{S}_{n+2}$ exists, but $\mathfrak{S}_{n+1} = 0$.

making $t = v + v'$,

$$\mathfrak{S}_t = \Sigma(x-h_{q_1})(x-h_{q_2}) \dots (x-h_{q_v})(x-\eta_{\xi_1})(x-\eta_{\xi_2}) \dots (x-\eta_{\xi_{v'}}) \frac{\left[\begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \end{matrix} \right] \times \left[\begin{matrix} h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{matrix} \right]}{\left[\begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{matrix} \right] \times \left[\begin{matrix} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{matrix} \right]} \quad \text{p. 471}$$

The conjugate form for which $t' = m + n - t$ and $m - v, n - v', vv'$ take the places of v, v' and $(m - v)(n - v')$, will be got by taking

$$\mathfrak{S}_{t'} = \Sigma(x-h_{q_{v+1}})(x-h_{q_{v+2}}) \dots (x-h_{q_m})(x-\eta_{\xi_{v'+1}})(x-\eta_{\xi_{v'+2}}) \dots (x-\eta_{\xi_n}) \frac{\left[\begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \end{matrix} \right] \times \left[\begin{matrix} h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{matrix} \right]}{\left[\begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{matrix} \right] \times \left[\begin{matrix} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{matrix} \right]}$$

which it will be perceived are identical, term for term, in the fractional constant factor, and differ only in the linear functions of x , which in $\mathfrak{S}_t$ and in $\mathfrak{S}_{t'}$ are complementary to one another. Our proper business is only with those forms for which $t < n$.

Art. (26). It will presently be seen to be necessary to ascertain the numerical relations between $\mathfrak{S}_{0,t}$ and $\mathfrak{S}_{1,t-1}$ when $t < n$, and this naturally brings under our notice the inquiry into the numerical relations which exist between the entire series of forms $\mathfrak{S}_{v,t-v}$ for a given value of t , corresponding to all values of v between 0 and t inclusive.

In order to avoid a somewhat oppressive complication of symbols, I shall take a particular numerical example, that is $m = 7, n = 6, t = 4$, and compare the values of $\mathfrak{S}_{0,4}, \mathfrak{S}_{1,3}, \mathfrak{S}_{2,2}, \mathfrak{S}_{3,1}, \mathfrak{S}_{4,0}$, all of which we know to be identical [to a numerical factor *près*] with one another and with the second simplified residue to f and ϕ , that being of the fourth degree in x ; our object in the subjoined investigation is to determine the numerical ratios of these several forms of $\mathfrak{S}$ to one another.

First, let $v = 0, v' = 4$. The leading coefficient $\mathfrak{S}_{0,4}$ is

$$\Sigma \frac{\left[\begin{matrix} \eta_5 \eta_6 \\ h_1 h_2 h_3 h_4 h_5 h_6 h_7 \end{matrix} \right]}{\left[\begin{matrix} \eta_5 \eta_6 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{matrix} \right]},$$

which we know *à priori* (it should be observed) to be essentially an *integral* function of the h and the η system. In this, the term containing η_6^3 will be evidently

$$\Sigma \frac{\left[\begin{matrix} \eta_5 \\ h_1 h_2 h_3 h_4 h_5 h_6 h_7 \end{matrix} \right]}{\left[\begin{matrix} \eta_5 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{matrix} \right]}, \quad (\text{A})$$

the η system to which the latter summation relates being now reduced to consist of $\eta_1, \eta_2, \eta_3, \eta_4, \eta_5$. In this expression, again, the coefficient of η_5^3 is evidently 1. Hence, therefore, the leading coefficient in $\mathfrak{S}_{0,4}$ contains the term $\eta_5^3 \eta_6^3$. p. 472

Secondly, let $v = 1, v' = 3$. The leading coefficient in $\mathfrak{S}_{1,3}$ becomes

$$\Sigma \frac{\begin{bmatrix} \eta_1 \eta_2 \eta_3 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \eta_5 \eta_6 \\ h_2 h_3 h_4 h_5 h_6 h_7 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 h_4 h_5 h_6 h_7 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \eta_5 \eta_6 \\ \eta_1 \eta_2 \eta_3 \end{bmatrix}}.$$

In this, the factor affecting η_6^3 will be

$$\Sigma \frac{\begin{bmatrix} \eta_1 \eta_2 \eta_3 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \eta_5 \\ h_2 h_3 h_4 h_5 h_6 h_7 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 h_4 h_5 h_6 h_7 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \eta_5 \\ \eta_1 \eta_2 \eta_3 \end{bmatrix}},$$

η_6 being now understood to be eliminated out of the η system included within the above summation. Again, in this latter sum the factor affecting η_5^3 will be

$$\Sigma \frac{\begin{bmatrix} \eta_1 \eta_2 \eta_3 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \\ h_2 h_3 h_4 h_5 h_6 h_7 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 h_4 h_5 h_6 h_7 \\ h_1 \end{bmatrix} \times \begin{bmatrix} \eta_4 \\ \eta_1 \eta_2 \eta_3 \end{bmatrix}}. \quad (\text{B})$$

η_5 and η_6 being now both eliminated out of the η system. This last sum can of course only represent a numerical quantity.

So in like manner, again, if $v = 2, v' = 2$, the coefficient of $\eta_6^3 \eta_5^3$ in $\mathfrak{S}_{2,2}$ will be similarly reducible to the form

$$\Sigma \frac{\begin{bmatrix} \eta_1 \eta_2 \\ h_1 h_2 \end{bmatrix} \times \begin{bmatrix} \eta_3 \eta_4 \\ h_3 h_4 h_5 h_6 h_7 \end{bmatrix}}{\begin{bmatrix} h_3 h_4 h_5 h_6 h_7 \\ h_1 h_2 \end{bmatrix} \times \begin{bmatrix} \eta_3 \eta_4 \\ \eta_1 \eta_2 \end{bmatrix}}. \quad (\text{C})$$

So, again, when $v = 3, v' = 1$, the coefficient of $\eta_6^3 \eta_5^3$ in $\mathfrak{S}_{3,1}$ will be

$$\Sigma \frac{\begin{bmatrix} \eta_1 \\ h_1 h_2 h_3 \end{bmatrix} \times \begin{bmatrix} \eta_2 \eta_3 \eta_4 \\ h_4 h_5 h_6 h_7 \end{bmatrix}}{\begin{bmatrix} h_4 h_5 h_6 h_7 \\ h_1 h_2 h_3 \end{bmatrix} \times \begin{bmatrix} \eta_2 \eta_3 \eta_4 \\ \eta_1 \end{bmatrix}}; \quad (\text{D})$$

and finally, the coefficient of $\eta_6^3 \eta_5^3$ in $\mathfrak{S}_{0,4}$ will be

$$\Sigma \frac{\left[\begin{array}{c} \eta_1 \eta_2 \eta_3 \eta_4 \\ h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_5 h_6 h_7 \\ h_1 h_2 h_3 h_4 \end{array} \right]}, \quad (\text{E})$$

out of all which sums it is to be remembered that η_5 and η_6 are supposed excluded from appearing. All these several coefficients being numbers in disguise, we may determine them by giving any values at pleasure to the terms in the h and η system.

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Let now $\eta_1 = h_1$, $\eta_2 = h_2$, $\eta_3 = h_3$, $\eta_4 = h_4$, then in (B) it will readily be seen that all the terms included within the sign of summation vanish identically, except the following, namely,

$$\frac{\left[\begin{array}{c} \eta_1 \eta_2 \eta_3 \\ h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_4 \\ h_1 h_2 h_3 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_2 h_3 h_5 h_6 h_7 \\ h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_4 \\ \eta_1 \eta_2 \eta_3 \end{array} \right]},$$

$$\frac{\left[\begin{array}{c} \eta_1 \eta_2 \eta_4 \\ h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_3 \\ h_1 h_2 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_2 h_4 h_5 h_6 h_7 \\ h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_3 \\ \eta_1 \eta_2 \eta_4 \end{array} \right]},$$

$$\frac{\left[\begin{array}{c} \eta_1 \eta_3 \eta_4 \\ h_2 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \\ h_1 h_3 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_3 h_4 h_5 h_6 h_7 \\ h_2 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \\ \eta_1 \eta_3 \eta_4 \end{array} \right]},$$

$$\frac{\left[\begin{array}{c} \eta_2 \eta_3 \eta_4 \\ h_1 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \\ h_2 h_3 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_2 h_3 h_4 h_5 h_6 h_7 \\ h_1 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \\ \eta_2 \eta_3 \eta_4 \end{array} \right]}.$$

In each of these expressions the first factor of the numerator is identical in value (by reason of the equations $h_1 = \eta_1$, $h_2 = \eta_2$, $h_3 = \eta_3$, $h_4 = \eta_4$) with $(-)^3 \times$ the second factor of the denominator, and the second factor of the numerator with $(-)^6 \times$ the first factor of the denominator; hence the coefficient of $\eta_5^3 \eta_6^3$ in $\mathfrak{S}_{1,3}$ is -4 .

In like manner the only effective terms of $\mathfrak{S}_{2,2}$ will be

$$\frac{\left[\begin{array}{c} \eta_1 \eta_2 \\ h_3 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_3 \eta_4 \\ h_1 h_2 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_2 h_5 h_6 h_7 \\ h_3 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_3 \eta_4 \\ \eta_1 \eta_2 \end{array} \right]}, \quad \frac{\left[\begin{array}{c} \eta_3 \eta_4 \\ h_1 h_2 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_2 \\ h_3 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_3 h_4 h_5 h_6 h_7 \\ h_1 h_2 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_2 \\ \eta_3 \eta_4 \end{array} \right]},$$

$$\frac{\left[\begin{array}{c} \eta_1 \eta_3 \\ h_2 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \eta_4 \\ h_1 h_3 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_3 h_5 h_6 h_7 \\ h_2 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \eta_4 \\ \eta_1 \eta_3 \end{array} \right]}, \quad \frac{\left[\begin{array}{c} \eta_2 \eta_4 \\ h_1 h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_3 \\ h_2 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_2 h_4 h_5 h_6 h_7 \\ h_1 h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_3 \\ \eta_2 \eta_4 \end{array} \right]},$$

$$\frac{\left[\begin{array}{c} \eta_1 \eta_4 \\ h_2 h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \eta_3 \\ h_1 h_4 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_1 h_4 h_5 h_6 h_7 \\ h_2 h_3 \end{array} \right] \times \left[\begin{array}{c} \eta_2 \eta_3 \\ \eta_1 \eta_4 \end{array} \right]}, \quad \frac{\left[\begin{array}{c} \eta_2 \eta_3 \\ h_1 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_4 \\ h_2 h_3 h_5 h_6 h_7 \end{array} \right]}{\left[\begin{array}{c} h_2 h_3 h_5 h_6 h_7 \\ h_1 h_4 \end{array} \right] \times \left[\begin{array}{c} \eta_1 \eta_4 \\ \eta_2 \eta_3 \end{array} \right]}.$$

Any other term will necessarily contain in the numerator a factor, whose symbolical representation will contain one of the quantities $\eta_1, \eta_2, \eta_3, \eta_4$, in the upper line, and one of the quantities h_1, h_2, h_3, h_4 , having the same subscript

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index, in the lower line, and which will therefore vanish; the number of effective terms being evidently the number of ways in which four things can be combined 2 and 2 together, and the value of each term is evidently $(-)^2(-1)^2 1$, so that the entire value of the coefficient of $\eta_5^3 \eta_6^3$ in $\mathfrak{S}_{2,2}$ is +6.

Precisely in the same manner, we shall find that the leading coefficient in $\mathfrak{S}_{3,1}$ will contain the term $-4\eta_5^3 \eta_6^3$, the (-1) resulting from the operation $(-1)^{13}(-)^{34}$, and in $\mathfrak{S}_{4,0}$ the term $+\eta_5^3 \eta_6^3$, the +1 resulting from the operation $(-1)^{43}$. Hence it appears that $\mathfrak{S}_{0,4}$; $\mathfrak{S}_{1,3}$; $\mathfrak{S}_{2,2}$; $\mathfrak{S}_{3,1}$; $\mathfrak{S}_{4,0}$ are to one another in the ratios of 1; -4; 6; -4; 1; and so in general for any values of m, n, t (t being less than m and less than n) it will be found that

$$\mathfrak{S}_{0,t}, \quad \mathfrak{S}_{1,t-1}, \quad \mathfrak{S}_{2,t-2} \dots \mathfrak{S}_{t,0}$$

will be in the ratios of the numbers

$$1; quad (-1)^{m-t} t; \quad (-1)^{2(m-t)} \frac{t(t-1)}{1 \cdot 2}; \quad (-1)^{3(m-t)} \frac{t(t-1)(t-2)}{1 \cdot 2 \cdot 3}; \quad \dots; \quad (-1)^{t(m-t)}.$$

Art. (27). The method employed in the preceding investigation will enable us to affix the proper sign and numerical factor to $\mathfrak{S}_{0,t}$, or in general to $\mathfrak{S}_{v,t-v}$, in order that it may represent the Bezoutian secondary of the degree t in x . This latter has been already identified with the simplified residue obtained by expanding $\frac{\phi x}{f x}$ under the form of an improper continued fraction. For this purpose, it will be sufficient to compare a single term of any such $\mathfrak{S}$ with the corresponding one in the Symmorphic Bezoutian secondary. Let us first suppose that $m = n$, f and ϕ being of the same degree. A glance at the form of the Bezoutian square will show that if we form the Bezoutian secondary of the degree $(n-i)$ in x , the coefficient of its leading term will contain the term $(-)^{\frac{(i-1)i}{2}}$; $(0, i)^i$; $(0, i)$ as usual denoting the product of the coefficient of x^n in f by the coefficient of x^{n-i} in ϕ , less the product of the coefficient of x^n in ϕ by that of x^{n-i} in f ; and as we suppose the first coefficients in f and ϕ to be each 1, if we term the other coefficients last spoken of a_i and α_i respectively, this said coefficient of the leading term of the i th Bezoutian secondary will contain the term $(-)^{\frac{(i-1)i}{2}} (\alpha_i - a_i)^i$, and consequently $(-1)^{\frac{(i-1)i}{2}} \alpha_i^i$ and $(-)^{\frac{i(i+1)}{2}} a_i^i$.

Now by the like reasoning to that employed in the preceding article, the coefficient of the leading term in $\mathfrak{S}_{m-i,0}$, that is

$$\Sigma(x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_1, \eta_2 \dots \eta_m \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}},$$

will contain the quantity $\Sigma(h_1 h_2 h_3 \dots h_i)^i$, and therefore will contain a term $\{\Sigma(h_1 h_2 h_3 \dots h_i)\}^i$, that is $(-)^i a_i^i$, which is equal to $(-)^i a_i^i$, since $(i-1)i$ is always even. Hence

$$\mathfrak{S}_{m-i,0} = (-)^{\frac{i-1}{2}} \times \text{the corresponding Bezoutian secondary.}$$

The above applies to the case where we have supposed $m = n$.

Art. (28). When this equality does not exist we may proceed as follows. Prefix to ϕx , the first coefficient of which is still supposed to be 1, a term εx^m , where ε is positive and indefinitely small, and let ϕx so augmented be called $\Phi(x)$. Then if $\eta_1, \eta_2 \dots \eta_n$ are the roots of ϕx , $\eta_1, \eta_2 \dots \eta_n$, together with the $(m-n)$ values of $\left(\frac{1}{\varepsilon}\right)^{\frac{1}{m-n}}$, will be the roots of $\Phi(x)$.

But it has already been proved that when (as here supposed) the first coefficient of $f x$ is 1, the Bezoutian secondaries to f and ϕ will be identical with those to f and Φ respectively; at least it has been proved that these latter, when $\varepsilon = 0$, but the form of Φ is preserved, become identical with the former, and consequently the same is true when ε is taken indefinitely small. Now if we call the $(m-n)$ roots of Φ which do not belong to ϕ , $\eta_{n+1}, \eta_{n+2} \dots \eta_m$, and make

$$\Psi_{m-i,0} = \Sigma(x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_1, \eta_2 \dots \eta_m \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}},$$

we have

$$\Psi_{m-i,0} = \Sigma P(h_{q_1}, h_{q_2} \dots h_{q_i}) \begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_{n+1}, \eta_{n+2} \dots \eta_m \end{bmatrix},$$

where

$$P(h_{q_1}, h_{q_2} \dots h_{q_i}) = (x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_1, \eta_2 \dots \eta_n \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}}.$$

But since $\eta_{n+1}, \eta_{n+2} \dots \eta_m$ are infinite in value,

$$\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_{n+1}, \eta_{n+2} \dots \eta_m \end{bmatrix} = \{(-\eta_{n+1})(-\eta_{n+2}) \dots (-\eta_m)\}^i \left(\frac{1}{\varepsilon}\right)^i.$$

Hence

$$\Psi_{m-i,0} = \left(\frac{1}{\varepsilon}\right)^i \Sigma P(h_{q_1}, h_{q_2} \dots h_{q_i}) = \left(\frac{1}{\varepsilon}\right)^i \mathfrak{S}_{n-i,0},$$

and

$$\mathfrak{S}_{n-i,0} = \varepsilon^i \Psi_{m-i,0}.$$

But by what has been shown antecedently, taking account of the fact of the